

Cloud Migration

What is Cloud Migration?

Workload is running in any cloud or onprem now we need to migrate.

Migration ku Krte hai?

Companies aapne crores paise lga kr server ab chor k cloud par ku ja rhe hai?

1: Cost Saving: (CapEX to Opex):

Phle server khardna pdta tha (Capital Expense)

cloud mai aap uthana hi bill dete hai jitna use kreng (Operational Expense)

2: Scalability (Elasticity)

3: Agility And Innovation

4: Disaster Recovery

Where To Migrate: 4 Ways

Public Cloud
Hybrid Cloud
Private Cloud
Multi Cloud

COLO Location

How many Strategies (The 6R's of Migration)

1: Rehosting (lift & Shift): (Code same rehga no changes in codebase) App jaise on-prem chal rha hai toh usse uthake kr hm Cloud mai Vm banke wha rkah dete hai.

Fastest, but cloud k features hm use nhi kr payenge.

2: Replat-forming (lift, tinker and shift): (Code same rehga), but database use kr rhe ho wo aapne cloud k db use kr liye manged wala (Azure SQL)

3: Refactoring And Rearchitecting: (Code base refactor hoga)

Sabse mahnga pd skta hai and lamba tarika, par long term mai sbse sasta aur scalable

4: Repurchasing (DROP/SHOP): Jo features phle cl rhi thi agr hm usse SAAS mai convert kr skte hai toh kr do (Local Email Server - Office 365)

5: Retiring: agr koi app aapka comapny mai use hi nhi ho rha hai toh usse migrate ku krna hai usse retire kro

6: retaining: agr koi app, hardware dependent app unne migrate mt kro jesa cl rhe hai wesi rkho.

How To Migrate?

Phase 1: Assessment And Discovery: Planning: Audit karo, Kidra data hai, kitni app hai, kon kon si app kis app se dependent hai, TOTAL COST OF OWNERSHIP (TCO), calculate karo cloud par kitna bill ayega.

Phase 2: Plan & design: Landing Zone Analysis (Basic Structure - N/W's, Subnet, IAM Security), APP k liye 6R's main ek strategy choose krni hogi.

Phase 3: Migrate & Validate: Execution: data aur app ko transfer kro, DMS (Database Migration Service), Cutover time

Phase 4: Optimize & Manage (Monitoring): Right Sizing, Cloud FinOps

Drawbacks:

- 1: Hidden Cost
- 2: Vendor Lock-in
- 3: Downtime
- 4: Security Risk During Transition

Solution:

- 1: FinOps
- 2: Docker Image/container/Terraform etc. (Open Source)
- 3: Blue Green Deployment
- 4: VPN, ExpressRoute